

CSE370 Spring 2026
Database Systems
Assignment: **01**
No of Questions: **03**
Total Marks: 20

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
 2. Your answer should be **handwritten**
 3. Submission Form: <https://forms.gle/9aTMkVyfwhLfn1aE8>
 4. **NO LATE SUBMISSION WILL BE ACCEPTED.**
-

Q1. [2+2 Marks] Select if the following tasks will require DDL(Data Definition Language) or DML(Data Manipulation Language).

Explain your answer briefly (at max 2 sentences) with mention to schema and state:

- a. Rename the column called “StudentID” into “SID”
 - b. Change the “class time” for section 7 to “01:00 PM - 2:05 PM”
-

Q2. [3+3 Marks] There are two types of data independence in database management systems:

Logical Data Independence and Physical Data Independence. Given the scenarios below, **identify** what type of **data independence** both are and **explain your answer briefly** with reference to the **Three-Schema Architecture**.

- You work for a social media company. The company has decided to move from cloud hosting to physical hardware to support the server due to increase in demand.
 - You work for a social media company. The company has decided to track the number of hours a user has stayed on their platform. This tracked_hours is not visible to the user but kept in the database.
-

Q3. [10 Marks] You are tasked with designing a Video Streaming System. The system must meet the following requirements:

- Each user will have a unique code, full name, email, and mobile number.
- Users can watch videos, which have a unique video ID, duration, upload_date, and play count.
- A video can be watched by many users and a user can watch many videos.
- Users can also comment on videos. Whenever a comment is made, the timestamp is recorded. A single user can comment multiple times.
- All videos must have comments, but not all users will leave a comment on the videos.
- Users can also follow other users.
- Some users might create a playlist. Each playlist has a playlist_name, tag, and a list of videos. Playlist_name is not unique, rather it is kept track of with the user's username and playlist_name combined.
- Users add videos to their playlist. One user can add many videos to their single playlist.

Now, draw an **ER diagram** of this system. Write down all assumptions (if any). It is mandatory to show the **cardinality ratio** and the **participation type** for each relationship.
